Supplementary Materials: Field Telemetry and Route Characterization of a Solar-Assisted Light Electric Vehicle

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Table S1. Estimated speed by session

Date	n	Mean (km/h)	95% CI	Median (km/h)	n_{eff}
25/07	114	7.74	6.97–8.84	7.31	19
27/07	369	14.85	12.49–18.85	14.38	33
28/07	448	16.02	13.41–19.99	15.16	30
29/07	587	14.67	11.91–18.56	12.73	28
30/07	638	14.68	12.22–18.07	13.52	33
31/07	504	15.63	13.92–18.35	15.55	37
11/08	173	8.70	7.59–9.73	9.36	33
12/08	196	8.00	7.41–9.65	8.38	57

95% CIs from a block bootstrap with blocks of 30 observations; n_{eff}: effective sample size accounting for first-order autocorrelation.

Table S2. Descriptive Spearman coefficients and empirical voltage slope by session

Date	$\rho(v, I)$	$\rho(v, \lvert I\rvert)$	R_{ef} (Ω)	95% CI	R^2
25/07	0.22	0.13	0.310	0.187–0.432	0.22
27/07	0.30	0.12	0.728	0.472–0.983	0.14
28/07	-0.09	0.06	0.206	0.143–0.269	0.10
29/07	0.06	0.07	0.193	0.129–0.256	0.09
30/07	0.04	0.08	0.234	0.119–0.348	0.05
31/07	-0.04	0.08	0.146	0.043–0.249	0.02
11/08	0.29	-0.15	0.316	0.138–0.495	0.15
12/08	0.43	-0.33	0.267	0.108–0.426	0.09

With eight sessions, the coefficients are descriptive and are not accompanied by hypothesis tests. The quantity R_{ef} is defined as $-\Delta V_{\text{bat}}/\Delta I_{\text{motor}}$ and represents an empirical slope, not an internal resistance.

Table S3. Session geometry (local coordinates)

Date	Path length (km)	Max. displacement (km)	N–S extent (km)	E–W extent (km)
25/07	1.226	0.10	0.07	0.10
27/07	7.610	1.25	1.39	0.59
28/07	9.966	1.25	1.38	0.83
29/07	11.957	1.24	1.39	0.52
30/07	13.009	1.24	1.39	0.53
31/07	10.944	1.25	1.39	0.53
11/08	2.091	0.06	0.06	0.11
12/08	2.178	0.13	0.05	0.13

Table S4. Records retained and excluded by date

Date	Retained	Excluded	% retained	Zero V_{bat}	Zero V_{panel}	Out-of-range I_{panel}
11/07	0	83	0.0	2	3	0
13/07	0	143	0.0	15	15	0
18/07	0	17	0.0	0	0	0
20/07	0	4	0.0	2	0	4
21/07	18	125	12.6	19	19	98
23/07	4	19	17.4	10	17	0
25/07	114	93	55.1	76	40	0
27/07	369	176	67.7	85	117	0
28/07	448	109	80.4	92	32	0
29/07	587	120	83.0	99	44	0
30/07	638	94	87.2	63	44	0
31/07	504	49	91.1	32	22	0
01/08	1	45	2.2	33	37	0
11/08	173	68	71.8	23	49	0
12/08	196	80	71.0	21	62	0

Categories are not mutually exclusive.

Table S5. Sensitivity to analytical decisions

Scenario	Sessions	n	Mean v (km/h)	Saturation (%)	Gross energy (Wh/km)	$R^2(v, \lvert P \rvert)$
Minimum coverage 50	8	3029	12.54	6.9	12.96	0.0026
Minimum coverage 100 (baseline)	8	3029	12.54	6.9	12.96	0.0026
Minimum coverage 200	5	2546	15.17	8.2	10.13	0.0028
Minimum coverage 400	4	2177	15.25	9.5	11.08	0.0024
Interval cap 10 s	–	–	–	–	11.59	–
Interval cap 30 s	–	–	–	–	12.96	–
Interval cap 60 s	–	–	–	–	15.25	–
Interval cap 120 s	–	–	–	–	18.84	–
Minimum speed 1 km/h	–	3029	12.54	6.9	12.96	–
Minimum speed 3 km/h	–	2667	14.05	7.3	11.53	–
Minimum speed 5 km/h	–	2397	15.20	7.6	10.44	–
Excluding saturated observations	–	2819	–	–	12.37	0.0004

"–" indicates that the scenario does not recalculate that quantity.